

Hyunseok Seung
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Professional Appointments

- 2025 – Present **Postdoctoral Research Associate**, *Department of Statistics, University of Wisconsin–Madison*, Madison, WI
Advisor: [Matthias Katzfuss](#)
- 2018 **Data Scientist**, *SK hynix Inc.*, Icheon, South Korea

Education

- 2025 **Ph.D.** in Statistics, *University of Georgia*, Athens, GA
Advisors: [Jaewoo Lee](#) and [Yuan Ke](#)
- 2018 **M.A.** in Applied Statistics, *Yonsei University*, Seoul, South Korea
Advisor: [Sangun Park](#)
- 2016 **B.A.** in Applied Statistics, *Yonsei University*, Seoul, South Korea

Research Interests

Scalable probabilistic machine learning with gradients and curvature:

- Scalable Gaussian processes with gradient observations
- Bayesian optimization and sample-efficient decision making
- Memory-efficient fine-tuning of large language models
- Curvature-aware optimization for deep neural networks

Publications

Preprints

- R1. **Hyunseok Seung** & Katzfuss, M. *Derivative Gaussian Processes on a Two-Direction Budget* [[arXiv](#)]. 2026.

Conference Proceedings

- C1. **Hyunseok Seung** & Katzfuss, M. *Scalable Derivative Gaussian Processes via Exact Gradient Reduction* in *Advances in Neural Information Processing Systems* (Accepted) [[arXiv](#)] (2026).
- C2. **Hyunseok Seung**, Lee, J. & Ko, H. *Low-Rank Curvature for Zeroth-Order Optimization in LLM Fine-Tuning* in *AAAI Conference on Artificial Intelligence* [[DOI](#)] (2026).
- C3. **Hyunseok Seung**, Lee, J. & Ko, H. *MAC: An Efficient Gradient Preconditioning using Mean Activation Approximated Curvature* in *IEEE International Conference on Data Mining* [[DOI](#)] (2025).

- C4. **Hyunseok Seung**, Lee, J. & Ko, H. *NysAct: A Scalable Preconditioned Gradient Descent using Nystrom Approximation* in *IEEE International Conference on Big Data* [DOI] (2024).
- C5. **Hyunseok Seung**, Lee, J. & Ko, H. *An Adaptive Method Stabilizing Activations for Enhanced Generalization* in *IEEE International Conference on Data Mining Workshop* [DOI] (2024).

Journal Articles

- J1. **Hyunseok Seung**, Han, K., Shen, Y. & Ke, Y. Enhancing COVID-19 Mortality Prediction with Online Autocovariance Change Points Detection. *Stat* **15**. [DOI] (2026).
- J2. **Hyunseok Seung**, Lee, J. & Ko, H. Mean Activation Curvature for Scalable Second-Order Optimization in Deep Networks. *Knowledge and Information Systems*. [DOI] (2026).
- J3. **Hyunseok Seung** & Park, S. Modified Likelihood Ratio Tests for Extreme Value Distributions. *Communications in Statistics–Theory and Methods* **52**. [DOI], 5742–5751 (2021).

Research Experience

Core Research Areas

- 2025 – Present **Postdoctoral Researcher**, *Department of Statistics, University of Wisconsin–Madison*
Scalable Gaussian Processes with Derivatives (advised by [Matthias Katzfuss](#))
 - Developed an exact gradient reduction principle for derivative Gaussian processes that reduces md gradient coordinates to at most m^2 target-specific directional derivatives while preserving the full-gradient conditional distribution for function-value prediction within each Vecchia factor. The reduction removes input dimension from dense Cholesky factorization, enabling scalable inference with gradient observations in high-dimensional settings.
 - Developed a two-direction approximation that further reduces md gradient coordinates to at most $2m$ directional derivatives, lowering per-target factorization cost from $\mathcal{O}(m^6)$ to $\mathcal{O}(m^3)$. In GP simulations, larger conditioning sets yield lower error than exact gradient reduction at 1/58 of the prediction time and 4.3% of the memory. The method also outperforms function-only baselines on high-dimensional BO tasks, including an 800-dimensional benchmark.
- 2022 – 2025 **Research Assistant**, *School of Computing, University of Georgia*
Curvature-Aware Optimization for Deep Learning (advised by [Jaewoo Lee](#))
 - Developed a curvature-aware zeroth-order optimizer for LLM fine-tuning that recasts preconditioning as learning a structured perturbation covariance approximating the inverse Hessian and incorporates RLOO variance reduction. The optimizer accelerates convergence and improves test accuracy while reducing peak memory by up to 27.3% relative to MeZO-Adam.
 - Developed a scalable curvature preconditioner for neural network training that replaces full activation covariance matrices with rank-one mean-activation statistics. Compared with KFAC, the method reduces end-to-end training time by up to 55.4% and improves vision transformer test accuracy by up to 3.6 percentage points, while maintaining memory usage comparable to SGD.

Collaborative and Earlier Research

- 2025 – Present **Postdoctoral Researcher**, *Department of Statistics, University of Wisconsin–Madison*

- Hyperspectral Foundation Models** (collaboration with [Townsend Lab](#))

 - Developing hyperspectral foundation models by pre-training vision transformers on hyperspectral data and fine-tuning them for downstream trait prediction, with conformal prediction used for uncertainty quantification.
- 2023 – 2024 **Research Assistant**, *Department of Educational Psychology, University of Georgia*
Topic Modeling (collaboration with [NPDA Lab](#))

 - Analyzed video and text data using automatic speech recognition and topic modeling, collaborating with researchers in mathematics education and psychology.
- 2021 – 2023 **Research Assistant**, *Department of Statistics, University of Georgia*
Time Series Forecasting (advised by [Yuan Ke](#))

 - Developed hybrid COVID-19 mortality forecasting models using online autocovariance change point detection, improving prediction accuracy by 6% and reducing training time by 99% compared to standard rolling-window cross validation.
- 2018 **Data Scientist**, *SK hynix Inc.*
Wafer Failure Early Detection System (supervised by [Sangun Park](#))

 - Identified key predictors of wafer failure using statistical models for high-dimensional semiconductor fabrication data.
- 2017 – 2018 **Research Assistant**, *Department of Applied Statistics, Yonsei University*
Modified Likelihood Ratio Tests (advised by [Sangun Park](#))

 - Developed modified likelihood ratio test statistics tailored to highly skewed settings to improve sensitivity to tail departures.

Talks and Presentations

Oral

- T1. *Scalable Derivative Gaussian Processes via Exact Gradient Reduction* Institute for Foundations of Data Science (IFDS) Research Expo (Madison, WI, USA). Aug. 2026.
- T2. *An Efficient Gradient Preconditioning using Mean Activation Approximated Curvature* IEEE International Conference on Data Mining (ICDM) (Washington, DC, USA). Nov. 2025.
- T3. *A Scalable Preconditioned Gradient Descent using Nystrom Approximation* IEEE International Conference on Big Data (Washington, DC, USA). Dec. 2024.
- T4. *Modified Likelihood Ratio Tests for Extreme Value Distributions* The Korean Data and Information Science Society (KDISS) (Pukyong University, Busan, South Korea). May 2018.

Posters

- P1. *Low-Rank Curvature for Zeroth-Order Optimization in LLM Fine-Tuning* AAAI Conference on Artificial Intelligence (Singapore). Jan. 2026.
- P2. *A Scalable Preconditioned Gradient Descent using Nystrom Approximation* AI Research Day, Institute for Artificial Intelligence (Athens, GA, USA). Apr. 2025.
- P3. *An Adaptive Method Stabilizing Activations for Enhanced Generalization* AI Research Day, Institute for Artificial Intelligence (Athens, GA, USA). Apr. 2024.
- P4. *Modified Likelihood Ratio Tests for Extreme Value Distributions* The Korean Statistical Society (KSS) (Pusan National University, Busan, South Korea). May 2018.

Software

2026	LITE [Code] PyTorch implementation of derivative GP inference using two directions per observed gradient, enabling larger Vecchia conditioning sets and batched prediction for scalable regression and Bayesian optimization.
	TERA [Code] PyTorch implementation of exact gradient reduction tailored to each prediction target for scalable derivative GP regression and Bayesian optimization in high dimensions.
2025	LOREN [Code] PyTorch implementation of zeroth-order LLM fine-tuning using low-rank curvature information and RLOO variance reduction.
	MAC [Code] PyTorch implementation of curvature preconditioning based on rank-one mean activation statistics for CNNs and vision transformers, including attention-aware preconditioning.
	CP [Code] Reproducible pipeline for COVID-19 mortality forecasting that adapts to detected autocovariance change points at national and state levels.
2024	NysAct [Code] PyTorch implementation of Nyström approximations to activation covariance for scalable gradient preconditioning in CNNs and vision transformers.
	AdaAct [Code] PyTorch implementation of adaptive optimization using activation variance to stabilize learned representations and improve generalization.

Teaching Experience

2018	Lecturer, Instructor of Record, Yonsei University STAT1001 Introduction to Statistics Fall 2018
2019 – 2023	Teaching Assistant, University of Georgia <i>Graduate Courses</i> STAT6430 Design and Analysis of Experiments Spring 2023 STAT6315 Statistical Methods for Researchers Spring 2023 STAT8330 Advanced Statistical Applications and Computing Fall 2022 STAT6420 Applied Linear Models Fall 2022 STAT8440 Statistical Inference in Bioinformatics Fall 2020 STAT6210 Introduction to Statistical Methods Fall 2020 STAT6210 Statistical Methods Fall 2019 <i>Undergraduate Courses</i> STAT4230 Applied Regression Analysis Spring 2022 STAT2360 Programming and Data Literacy using R Fall 2021 STAT4210 Statistical Methods Spring 2021 STAT3110 Introduction to Statistics for Life Sciences Summer 2020 STAT3120 Introduction to Probability for Life Sciences Spring 2020
2017 – 2018	Teaching Assistant, Yonsei University

STAT1001 Introduction to Statistics
STAT1001 Introduction to Statistics
STAT1001 Introduction to Statistics

Spring 2018
Fall 2017
Spring 2017

Honors & Awards

2018 Industry-Sponsored Scholarship Recipient, *SK hynix Inc.*
2018 Best Paper Presentation Award, *Korean Data and Information Science Society*
2017 – 2018 Brain Korea 21 Fellowship, *Korean Government-Funded Graduate Program*
2015 Honors Award, *Yonsei University*

Academic Service

Reviewer

2026 International Conference on Artificial Intelligence and Statistics (AISTATS 2027)
 International Conference on Learning Representations (ICLR 2027)
 Journal of the American Statistical Association (JASA), Theory and Methods
 Journal of Computational and Graphical Statistics (JCGS)
 Advances in Neural Information Processing Systems (NeurIPS 2026)

Registered Student Organization Advisor

2026 – 2027 KoJobs UW–Madison, University of Wisconsin–Madison

Last updated: October 8, 2026